



Fig. 4: 3D-printed components of the humanoid robotic arm with part names and quantities

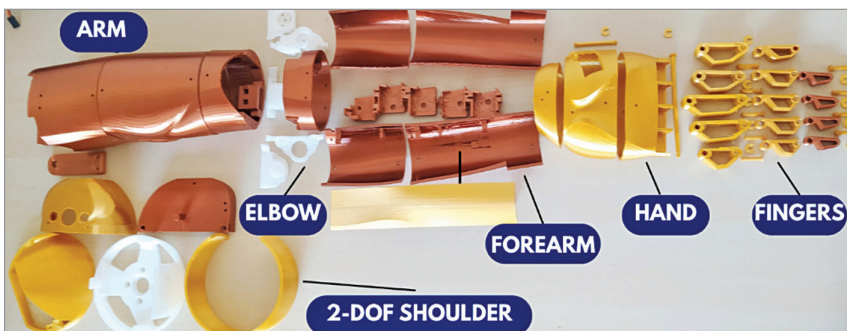


Fig. 5: Assembled and grouped 3D-printed sections of the humanoid robotic arm, including shoulder, elbow, forearm, hand, and fingers

assembly of the arm.

Assembly of the fingers and palm. After completing 3D printing, begin by assembling the fingers. Fix each phalanx segment using M3 screws to create movable joints, ensuring proper alignment, as shown in Fig. 6. Cut about 30cm of fishing nylon, tie a knot at one end, and pass it through the finger segments. This nylon serves as a

tendon for controlling finger motion. Pull the thread to check smooth bending and make adjustments if needed. Also ensure proper extension when released. Repeat the same steps for all five fingers.

The setup forms a 3D-printed robotic hand with articulated fingers capable of human-like motion. Segmented joints provide flexibility, while the fingertip sections improve

TABLE 3
3D PRINTING SETTINGS

Parameter	Value
Material	PLA
Layer height	0.1mm
Print speed	50mm/s
Shell thickness	0.8mm
Infill density	100%
Nozzle temperature	220°C
Bed temperature	50°C
Supports	Everywhere
Build plate adhesion	Raft
For nylon (optional):	
Nozzle temperature	255-300°C
Bed temperature	65-70°C

grip during object handling. The tendon-driven mechanism enables controlled curling and releasing of fingers. All fingers are mounted on a palm structure, which can house actuators such as servo motors, making the design suitable for robotics, prosthetics, and pick-and-place tasks.

Assembly of the arm. Next, assemble the arm structure. Use two long connectors and four M3 (8mm) screws to join the middle section of the hand to the upper section. Select appropriate connectors based on spacing between mounting holes. Fix the four fingers to the hand using a long axle and secure it with an M3 screw. Attach the thumb using the thumb axle and lock it with an axle clip. Pass all nylon threads through the holes at the base of the hand; tension can be adjusted later.

For the forearm, join the front and middle sections using screws. Mount the servo base with 12mm and 18mm M3 screws. Install the 25kg servo into its bracket and attach the servo arm to the elbow joint. Finally, connect the servo to the Arduino to calibrate its minimum and maximum positions for proper motion (see Fig. 7).

After that, mount the 25kg servo in its bracket (housing). Attach the servo arm to the left side of the forearm elbow and secure it with screws. Then connect the